

Smartphone-Based Human Activity Recognition Using Supervised Machine Learning

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1. Background, Motivation, and Significance

When our group first began exploring topics, the idea of Human Activity Recognition (HAR) immediately captured my interest. The motivation felt incredibly relevant and personal; in a world where smartphones are extensions of ourselves, the ability to use them to understand and quantify our physical lives is a powerful concept. As detailed in our final report, HAR isn't just a technical challenge—it has real-world significance in fields like mobile health, elder care, and fitness. I was motivated by the prospect of taking raw, abstract sensor data and transforming it into something meaningful and human: a label that says, "this person is walking," "sitting," or "climbing stairs."

My role from our very first meeting was to help shape the narrative and structure of our project. As our team's meeting minutes document, I was tasked with planning the report format and gathering the materials for our presentation. This role gave me a unique vantage point, allowing me to think about how we would communicate our journey and findings from the very beginning. I saw my contribution not just as executing tasks, but as helping to tell the story of our research in a clear and compelling way, ensuring that the significance of our work was front and center.

2. Research Questions and Data

Our project was guided by a core research question: "Can a supervised learning model accurately classify physical activity based on smartphone sensor features?" This question directly informed our approach to the UCI-HAR dataset. When we first examined the dataset, its complexity was striking over 10,000 samples, each with 561 features. It was a rich but intimidating collection of data from smartphone accelerometers and gyroscopes.

My initial contribution was to work with the team to understand and frame this data for our report. We had to break down its complexity for our audience, explaining the different activities and the nature of the sensor readings. A key part of our early process, which I helped document, was the decision to combine the provided training and testing sets. This allowed us to perform our own cross-validation, giving us a more robust and unbiased evaluation of our models later. This foundational step was critical, and ensuring it was clearly explained in our final report was one of my primary responsibilities.

3. Methodology: My Role in Preprocessing and Documentation

The most critical phase of the project, in my opinion, was the methodology, particularly our approach to data preprocessing. Confronted with 561 features, it was clear that dimensionality reduction was essential. I actively participated in the group discussions where we decided to implement Principal Component Analysis (PCA). My task was to take the complex results of the PCA analysis and translate them into the clear visuals and explanations seen in both our report and our final presentation. I focused on creating the narrative that explained *why* we chose to retain 95% of the variance and how that allowed us to drastically reduce the feature set without losing critical information. This wasn't just about running code; it was about making a strategic decision and justifying it.

Furthermore, I took the lead in ensuring our methodology was meticulously documented. This involved drafting sections of the report that detailed our use of StandardScaler for normalization and our 10-fold cross-validation strategy. I learned that a successful project isn't just about getting a good result; it's about creating a process that is transparent, repeatable, and easy for others to understand.

4. Modeling and Evaluation

As my teammate Shyam focused on the technical implementation and hyperparameter tuning of our three models—Random Forest, Support Vector Machine (SVM), and a Multilayer Perceptron (MLP)—my role shifted to analysis and communication. I worked closely with Krishnarjun and Ekshitha to interpret the performance metrics. We didn't just look at the final accuracy scores; we dove deep into the confusion matrices.

My contribution was to take these matrices and build a story around them. For example, I was the one who pointed out the recurring pattern of confusion between "Sitting" and "Standing" and framed it in our report as a logical and expected challenge, given the similarity in sensor data for static postures. This analytical insight was crucial for demonstrating a deeper understanding of our results. I created the model comparison charts for our presentation and report, ensuring they were not just data dumps but clear, comparative visuals that made our key finding—the superiority of the Random Forest model—immediately obvious.

5. Results and Discussion: A Personal Reflection

The results were incredibly gratifying. Seeing our Random Forest model achieve over 95% accuracy was a testament to our team's diligent work. However, my most significant learning came from the "Challenges & Learnings" section of our work. The struggle with potential sensor noise and activity overlap taught me that data science is an imperfect science. No model is perfect, and acknowledging its limitations is a sign of a thorough analysis.

I personally wrote and refined much of the discussion in our final report, focusing on the "why" behind our results. Why did Random Forest outperform the others? I argued it was due to its inherent ability to handle high-dimensional data and its resistance to overfitting, which made it a natural fit for our problem. This process of critical thinking and articulating our findings was where I felt I grew the most as a data scientist.

6. Conclusion and Future Works

In contributing to the conclusion and future works sections, I felt a sense of completion and foresight. Our project successfully demonstrated that supervised learning is a powerful tool for HAR. My journey through this project was one of learning by doing. I started with a focus on planning and formatting, but that role evolved into one of analysis, interpretation, and storytelling.

I learned that my strength lies in bridging the gap between the technical implementation and the final, human-readable insight. I am proud of my contributions, from shaping the initial outline to analyzing the final confusion matrix. This project has given me hands-on experience and, more importantly, the confidence to tackle complex data problems collaboratively and to communicate the results in a way that is both accurate and meaningful.